

LinguaQuest

Educational Mobile App

PROJECT TYPE Educational Mobile App	PLATFORM Cross-platform mobile app with web backend
AUDIENCE Grade 7 Filipino learners and teachers	PRIMARY FOCUS Gamified listening comprehension

Overview

A gamified listening comprehension mobile app for Grade 7 Filipino learners. Learners sail with Captain Salita and Dagat across seven themed islands of the Listening Sea, mastering one listening skill per island, collecting Shards of Understanding, and silencing Ingay - the Spirit of Noise. Built with Expo SDK 55, React Native 0.83, and a Next.js 16 + Prisma + Neon backend with real-time multiplayer powered by Pusher.

Core Features

- 7 story-based islands, each tied to a Grade 7 Filipino listening skill (vocabulary, pacing, main idea, tone, details, sequence, full application)
- Single-play audio discipline with skill-targeted hints and Captain-read explanations on correct answers
- Real-time multiplayer Crew Voyage: 5-round ship-repair battles with crew voting, repair phases, and in-room chat over Pusher channels
- Pin-based progression with 80% pass threshold per pin and per-pin accuracy tracking for student and teacher dashboards
- Voice-acted Captain Salita and Ingay clips, per-island music, and Cloudinary-hosted audio pipeline
- Shard collection, Alingawngaw Crystal assembly, and exportable Certificate of Completion via react-native-view-shot
- Role-based portals: STUDENT play, TEACHER analytics with mistake review, and ADMIN audio upload console
- JWT auth (Jose) with Google OAuth, server-side answer validation, and Expo SecureStore token storage

Experience and Design

- Seven themed islands map listening skills to a continuous story and voyage progression.
- Skill-targeted hints and explanations support vocabulary, pacing, main idea, tone, detail, and sequence.
- Crew Voyage multiplayer uses voting, ship-repair rounds, and room chat for cooperative practice.
- Shards, badges, crystals, and a completion certificate make mastery visible.

Architecture and Data

- Expo and React Native deliver the learner experience across supported mobile platforms.
- A Next.js backend exposes role-aware APIs backed by Prisma and Neon PostgreSQL.
- Pusher channels synchronize multiplayer rooms, voting, repair phases, and chat.
- Cloudinary delivers voice acting, island music, and other learning media.

Security and Privacy

- JOSE-based JWT authentication uses Expo SecureStore for device-side token protection.
- Google OAuth provides an additional controlled sign-in path.
- Student, teacher, and administrator permissions are enforced on backend routes.
- Correct answers are validated on the server and are not sent to the client before submission.
- The app requests no microphone access because learners listen and answer rather than record.

Technology Stack

Expo SDK 55	Application capability
React Native 0.83	Application framework and UI
TypeScript	Application language
NativeWind	Interface styling and design system
Zustand	State, data flow, and architecture
TanStack React Query	State, data flow, and architecture
Pusher	Real-time communication
Next.js 16	Application framework and UI
Prisma	Database access and schema management
Neon PostgreSQL	Authentication, database, or cloud data
Cloudinary	Media storage and delivery

Performance and Reliability

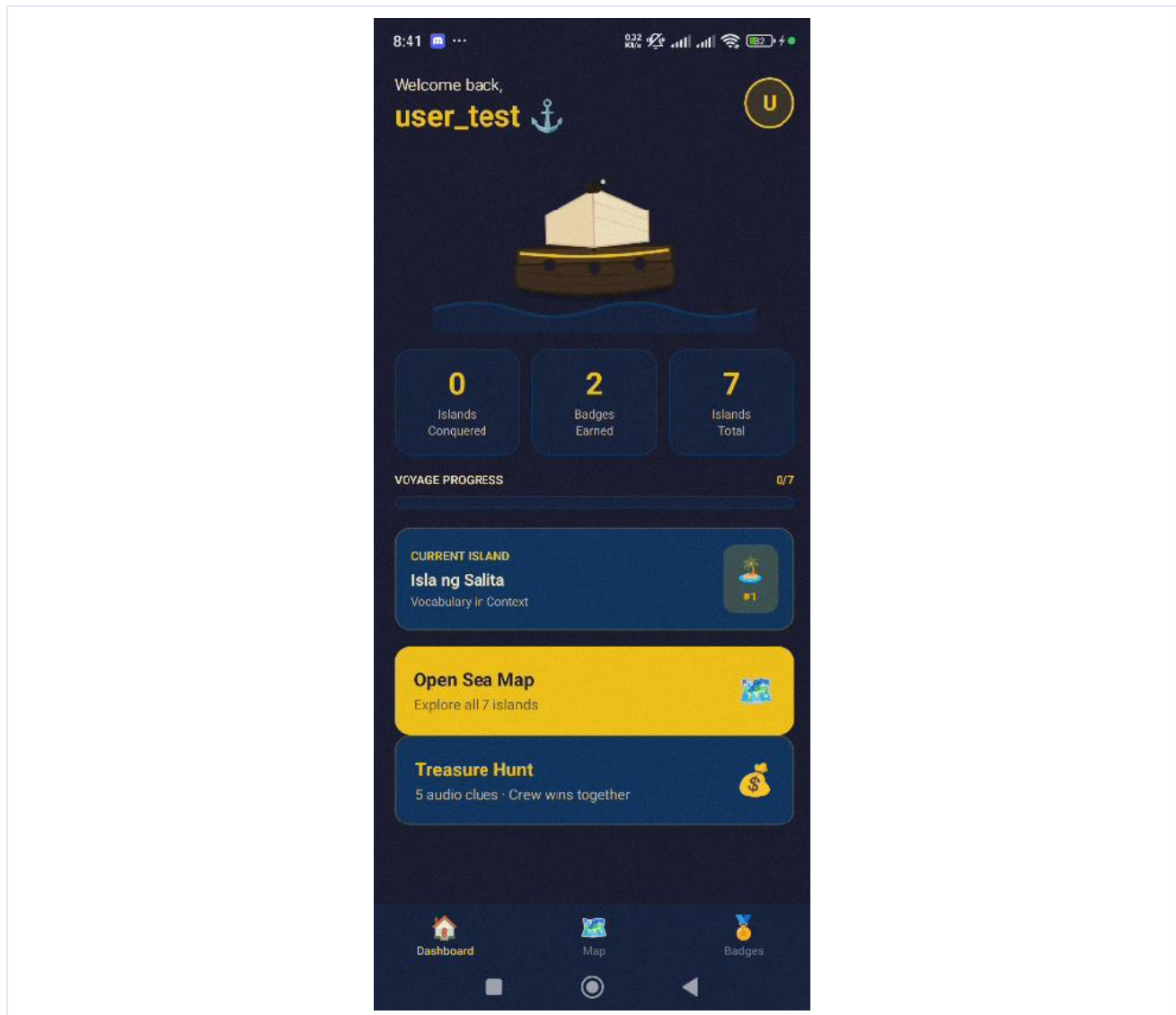
- TanStack React Query caches islands, pins, progress, and badges.
- Zustand provides lightweight local state for authentication, audio, and voyage progress.
- Room-scoped Pusher events limit multiplayer updates to active crews.
- Cloud-hosted audio avoids bundling the full media library into the application package.

Platform Requirements

- Supported Android or iOS device with audio output.
- Stable internet connection for synchronized progress and multiplayer rooms.
- Google account or application credentials when sign-in is required.
- Headphones are recommended for focused listening-comprehension activities.

PAGE OVERVIEW

Home - Voyage Progress

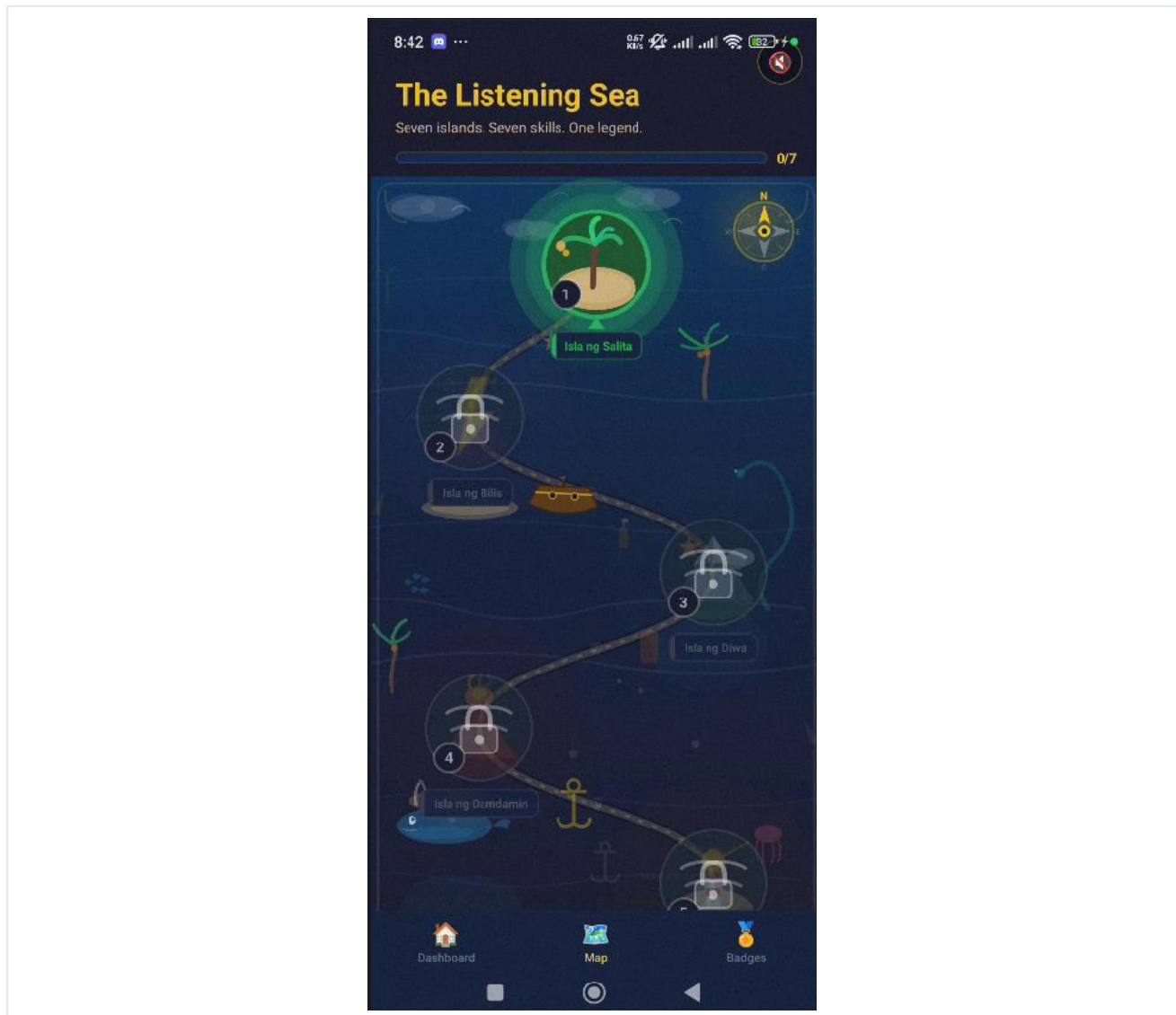


Home - Voyage Progress

Interface view from LinguaQuest.

PAGE OVERVIEW

Pirate Map - The Listening Sea

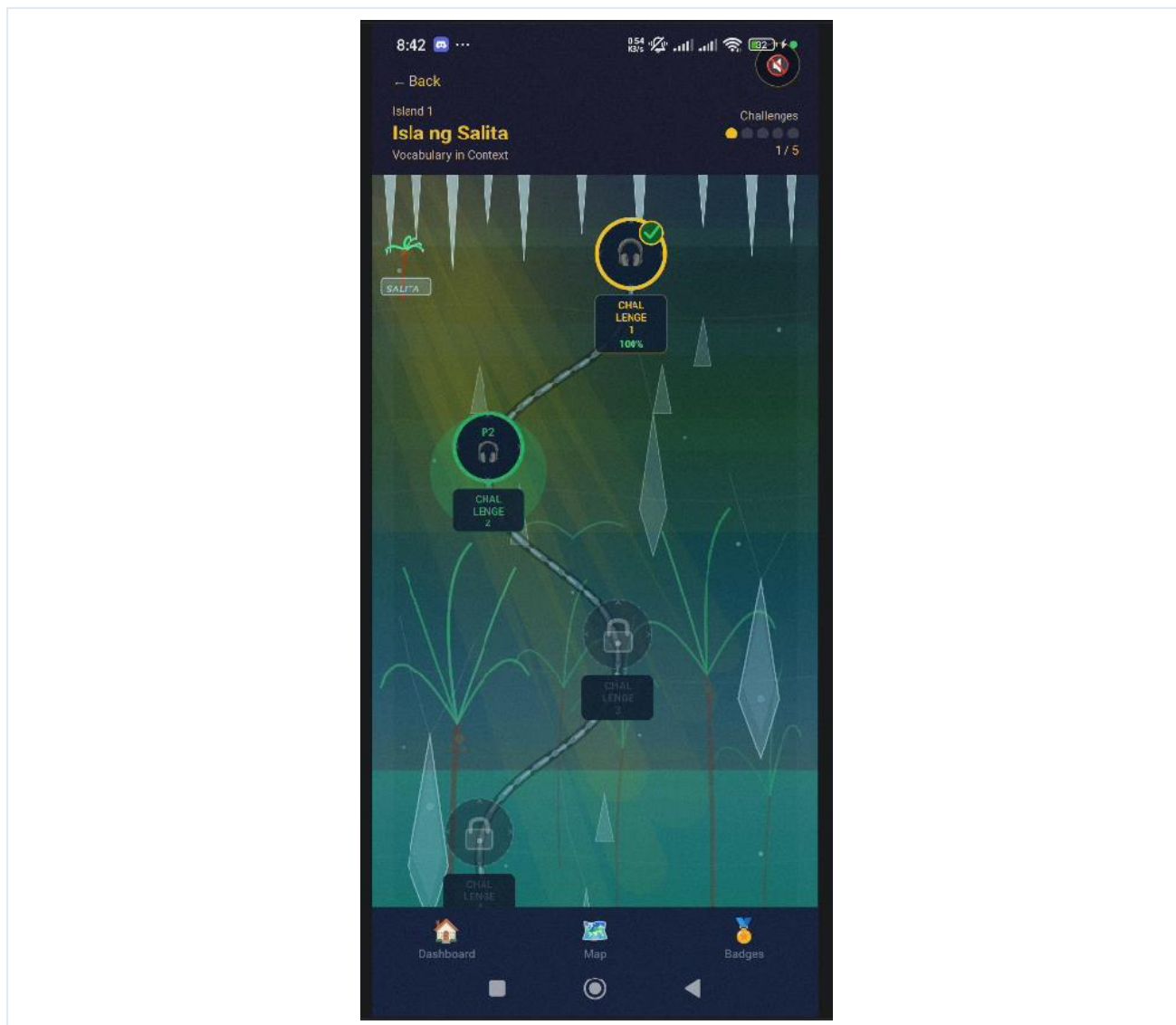


Pirate Map - The Listening Sea

Interface view from LinguaQuest.

PAGE OVERVIEW

Inside an Island

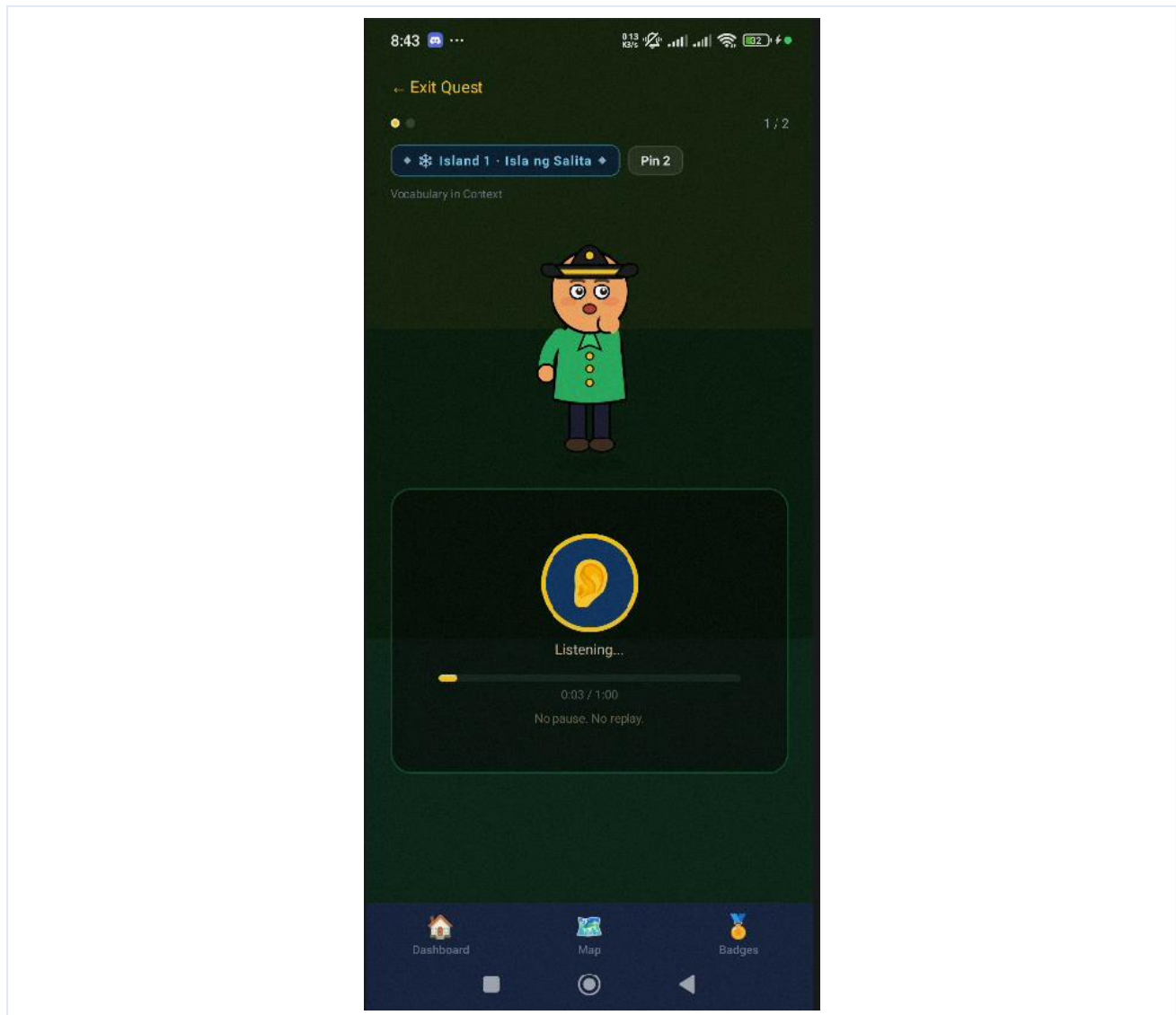


Inside an Island

Interface view from LinguaQuest.

PAGE OVERVIEW

Listening Challenge

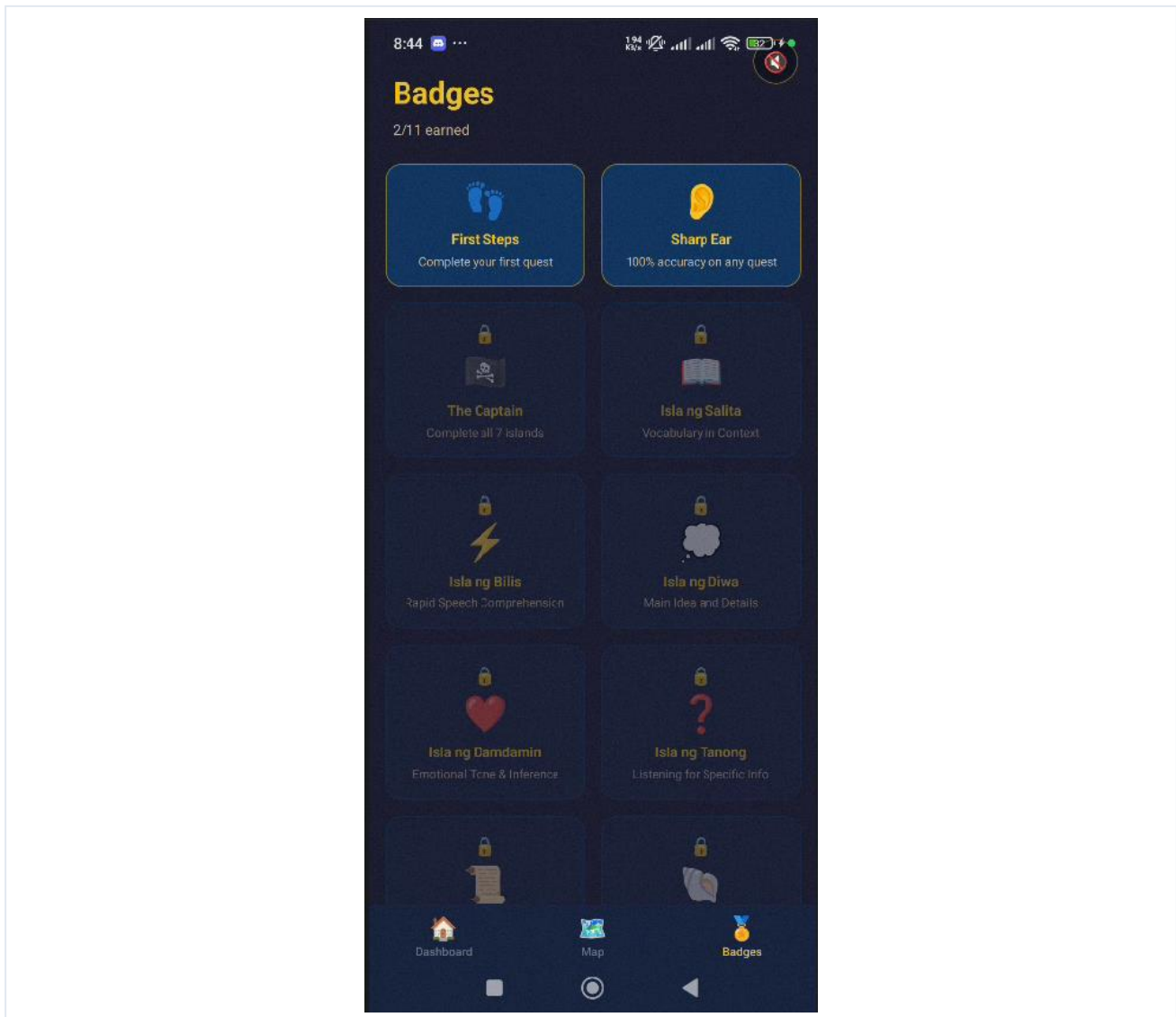


Listening Challenge

Interface view from LinguaQuest.

PAGE OVERVIEW

Badges & Achievements

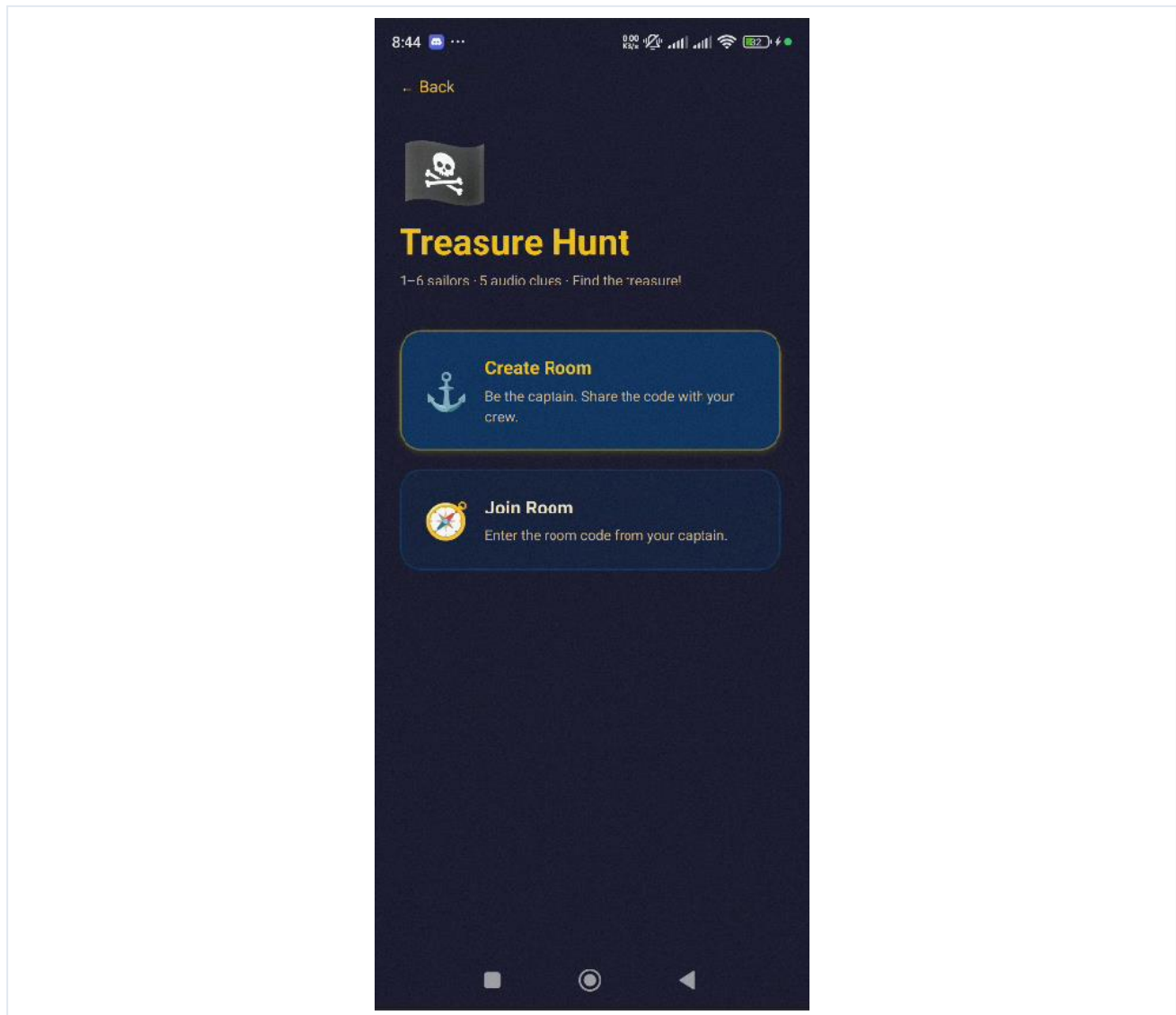


Badges & Achievements

Interface view from LinguaQuest.

PAGE OVERVIEW

Multiplayer Crew Lobby

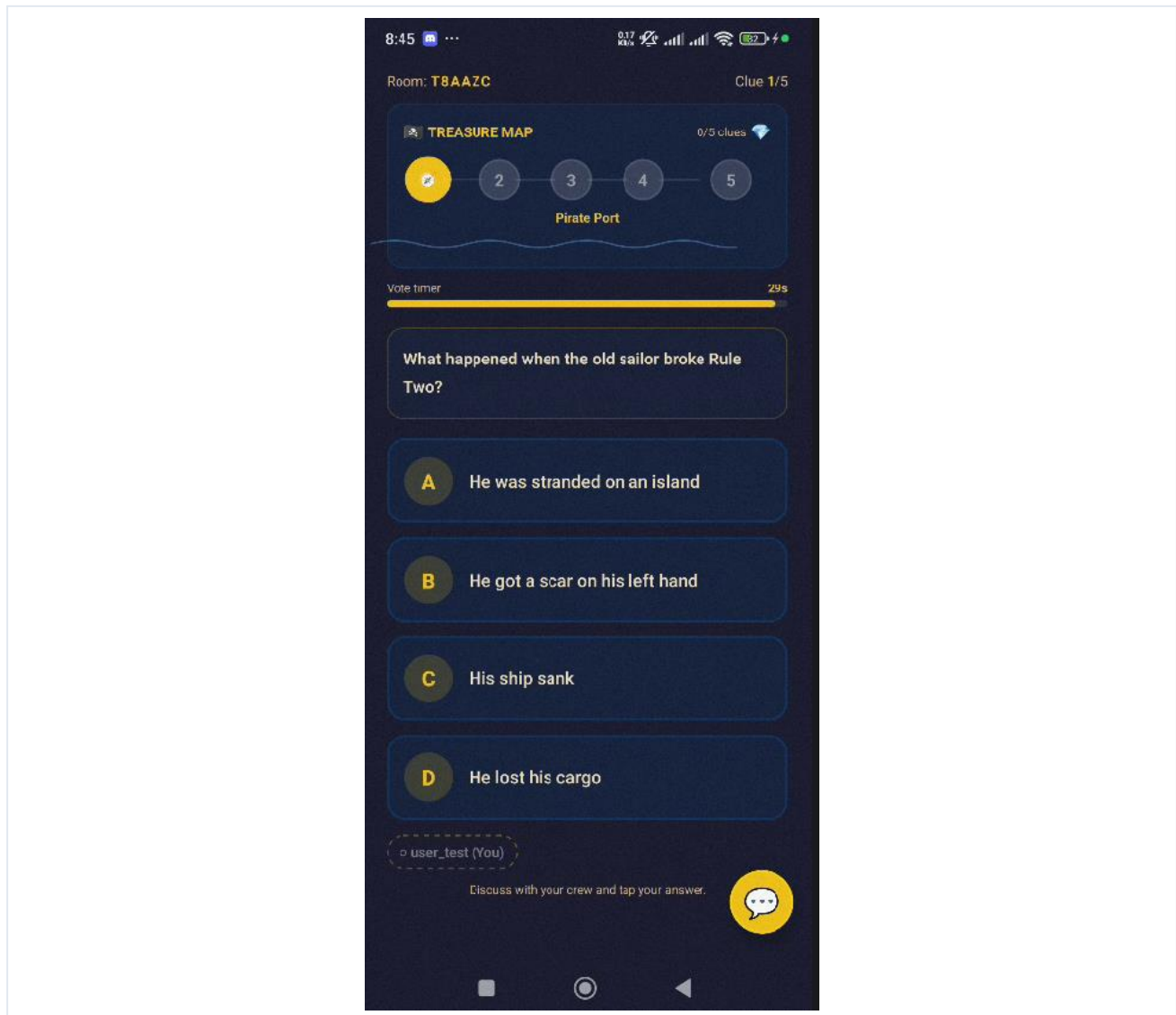


Multiplayer Crew Lobby

Interface view from LinguaQuest.

PAGE OVERVIEW

Inside Multiplayer - Treasure Map



Inside Multiplayer - Treasure Map

Interface view from LinguaQuest.